MILITARY SPECIFICATION

TOOLS AND TEST EQUIPMENT: 10559650 (LASER RANGEFINDER AN/VVS-1)

1. SCOPE

1.1 This specification covers one type of equipment known as Laser Rangefinder AN/VVS-1 Tools and Test Equipment: 10559650 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-4150	Carrying Case: 10559651
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment: Acquisition, Maintenance, and Disposition of
MIL-T-50606	Test Set, Receiver-Transmitter: 10559600

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Army, Frankford Arsenal

F10559650	Special Tools and Test Equipment - Laser Rangefinder AN/VVS-1 Assembly

PACKAGING DATA SHEET

10559650	Packaging of Special Tools and Test Equipment - Laser Rangefinder AN/VVS-1 Assembly: 10559650

Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

REQUIREMENTS

3.1 Fabrication.- The equipment shall be manufactured in accordance with Drawing F10559650 and drawings pertaining thereto. The following items shall each have met the requirements of their detailed specifications and drawings, respectively, prior to assembly.

Nomenclature	Drawing	Specification
Test Set, Receiver-Transmitter	10559600	MIL-T-50606
Carrying Case	10559651	MIL-C-4150
Cable, Test Set	10559663	
Extender Cable, PFN Charge PS	10559656	
Test Cable - 6A1/6A3	10559665	
Test Cable - 6A2/6A3	10559666	
Test Cable - 6A3/6A4	10559667	
Test Cable - 6A1/6A5	10559668	
Test Cable - 6A1/Computer	10559669	
Spanner Wrench	10559655	
Extension, Socket Wrench $\frac{1}{2}$ Drive, 18 Inches Long	10559656	
Extender Card, Counters	10559508	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

Nomenclature	Drawing	Specification
Extender Card, Line Driver	10559514	
Extender Card, LV Power Supply	10559515	
Extender Card, Reply Gating	10559516	
Extender Card, -1600 Volts Power Supply	10559517	
Extender Card, Select Logic	10559518	
Extender Card, Indicator, Digital Display	10559260	
Extender Card, Control, Gunner's, Laser Ranger	10559236	
Extender Card, Indicator, Command, Display	10559550	
Kit, Purging Adapter	10559660	

3.2 General specification.- The equipment shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence

(b) Dimensions and tolerances

(c) Part identification and marking.

3.2.1 Electronic workmanship.- The equipment shall meet the workmanship requirements of Drawing F10559650 and drawings pertaining thereto.

3.3 Performance.- Unless otherwise specified, equipment shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Environmental.

3.3.1.1 Storage temperature.- The equipment shall not be damaged nor shall its performance at ambient temperature (60°F to 90°F) be impaired after having been exposed to and thermally stabilized at temperatures of -65°F and +160°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

3.3.1.2 Vibration.- The equipment shall not be damaged nor shall its performance be impaired after exposure to the following vibratory conditions:

a.	Amplitude:	Simple harmonic motion at 0.03 inch peak (0.06 inch total excursion) maximum.
b.	Acceleration:	1.5g maximum.
c.	Sweep Time:	10 to 55 to 10 Hertz in 5 minutes, two sweeps in each of three mutually perpendicular axes.

3.3.1.3 Shock.- The equipment shall not be damaged nor shall its performance be impaired after exposure to a total of 6 shock impulses, one in each direction of the three mutually perpendicular axes. Each shock impulse shall be a sawtooth or half-sine wave with a duration of 11 milliseconds. The peak amplitude of each shock impulse shall be 30 g.

QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three sets of equipment. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection or sampling procedures herein and "Submission of Product" provisions of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and sub-
ssemblies shall be inspected in accordance with the inspection provisions
ntained in Supplementary Quality Assurance Provisions (SQAP) listed in
ıe technical data package (TDP). In the absence of SQAP's, the
pplicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection.- Examination and tests related
Section 3 herein shall be performed on a single defect (individual
ıaracteristic) basis in accordance with MIL-STD-105 and the sampling
ans specified in Tables I and II herein. Examination and test for
ckaging and marking shall be in accordance with MIL-P-14232 and Section 5
rein. The tabulated classification of defects in Tables I and II shall
nstitute the minimum inspection to be performed by the supplier prior
Government acceptance or rejection by item or lot.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened
r all defective characteristics. Removal or correction of defective
its and resubmittance of rejected lots shall be in accordance with
cceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One set of equipment shall be selected at random
a Government representative as a control sample from each 100 produced.
e control sample shall meet the requirements and tests in Table I.

TABLE I

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawings - visual
302. General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental.- Three sets of equipment shall be selected
random by a Government representative as a control sample from each
produced or from each month's production, whichever occurs first. The
ntrol samples shall meet the requirements and tests in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
303. Storage temperature	3.3.1.1	4.6.2.1
304. Vibration	3.3.1.2	4.6.2.2
305. Shock	3.3.1.3	4.6.2.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" provisions of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Test conditions.- Unless otherwise specified, the test conditions shall be in accordance with the "Test Facilities" provisions of MIL-F-13926.

4.6.1.1 Initial conditions.- All items of equipment shall be packed in the carrying case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

4.6.2 Environmental tests.

4.6.2.1 Temperature.- The test apparatus shall be in accordance
the applicable "Test Facilities" provisions of MIL-F-13926. Per-
this test in accordance with Procedure I of MIL-F-13926, except
the low storage temperature shall be -65°F, in lieu of -80°F, the
rmediate operating conditions shall be omitted, and the rate of
erature change shall not be greater than 7°F per minute. Following
erature cycling and stabilization at ambient temperature, examine
carrying case and its contents for damage to determine compliance
3.3.1.1. Perform the "MODE SELECT switch" test and "RCVR-XMTR
US meter reading" test of MIL-T-50606 on the Receiver-Transmitter
Set and the electrical tests specified in the individual specifi-
ons or drawings (see 3.1), as applicable, on the remaining components
cables to determine compliance with 3.3.1.1.

4.6.2.2 Vibration.- Perform this test utilizing the apparatus
ified in Method 514.1 of MIL-STD-810 and in accordance with the
atory conditions specified in 3.3.1.2. Following vibration
sure, examine the carrying case and contents for damage to deter-
compliance with 3.3.1.2. Perform the "MODE SELECT switch" test and
R-XMTR STATUS meter reading" test of MIL-T-50606 on the Receiver-
smitter Test Set and the electrical tests specified in the individual
ifications or drawings (see 3.1), as applicable, on the remaining
onents and cables to determine compliance with 3.3.1.2.

4.6.2.3 Shock.- Perform this test utilizing apparatus specified
ethod 516.1 of MIL-STD-810, in accordance with Procedure I therein,
pt that the conditions of 3.3.1.3 shall apply. Following exposure
hock, examine the carrying case and contents for damage to determine
liance with 3.3.1.3. Perform the "MODE SELECT switch" test and
R-XMTR STATUS meter reading" test of MIL-T-50606 on the Receiver-
smitter Test Set and the electrical tests specified in the indivi-
specifications or drawings (see 3.1), as applicable, on the
ining components and cables to determine compliance with 3.3.1.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10559650. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The equipment is intended to be used for maintenance of the Laser Rangefinder AN/VVS-1.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable packaging data sheet number (see 5.1).

(c) Selection of applicable levels of preservation, packaging and packing.

(d) Applicable stock number.

(e) Requirement for First Article Submission.

Custodian:
Army - MU

Preparing Activity
Army - MU
Project No. 4931-A053

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-J4000
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:30Z
Check the source to verify that this is the current version before use.